CrewForge

PS1 · Race-proof team formation at scale — Hack4Hack Crew Hiring Challenge 2026

Objective: Database-enforced team membership for a 10,000-participant hackathon. Prevent over-capacity teams, duplicate membership and unsafe retries when many users compete for one seat.

Problem

Concurrent requests for the final team seat can overfill teams or create duplicate memberships. Client-side checks cannot protect state across app instances.

Data model

MongoDB: users, teams, memberships, joinRequests, invites, audit, rateLimits, securityEvents and authTokens. Unique compound indexes enforce one membership per participant per hackathon and one pending request per participant.

Realtime & lifecycle

MongoDB change streams feed bounded SSE updates. Invites are high-entropy, expiring, single-use tokens stored as hashes. Leader departure transfers leadership; final-member departure closes the team.

Race prevention

Authenticate and validate; create a pending request; leader approves inside one MongoDB transaction. The transaction inserts membership, conditionally increments memberCount only below maxSize, marks the request accepted and appends an audit event. Any failed guard aborts the transaction.

Concurrency proof

50 pending requests and 50 concurrent leader approvals compete for one remaining seat. The replica-set integration test asserts exactly 1 accepted, 49 rejected, 2 memberships including the leader, memberCount = 2 and one acceptance audit event.

Security

HttpOnly sessions, server-side authorization, same-origin checks, persistent rate limits, input validation, hashed auth tokens and transactional audit events.

Measured evidence

65 / 65	20.92 ms	100 VUs	0%
local tests passed	CI p95 latency	3,000 requests	HTTP failures

CI verified 10,000 seeded participants, 2,000 teams and 10,000 membership records. All 65 tests passed: 45 API contract, 9 unit and 11 MongoDB integration. Load test: 100 VUs, 3,000 requests, 97.43 req/s, 20.92 ms p95, 0% HTTP failures and 6,000/6,000 board checks. GitHub runner + disposable MongoDB replica set; regression benchmark, not a production-region measurement. CI: 41873925487 · CodeQL: 41873925485.

Trade-offs & next week

Transactions and unique indexes keep invariants next to the source of truth but require a replica set and add transaction/pool overhead. SSE is simpler than WebSockets for one-way invalidation, but change streams consume connections. Next: complete Atlas and verified sender configuration, deploy the full-stack preview, run invite/auth/rate-limit/SSE staging checks, and measure multi-session behavior under controlled load.

Preview scope: the public URL is an interactive frontend preview with browser-local state. Server-side API contracts, MongoDB transaction behavior and concurrency correctness are independently verified in CI. The benchmark is not a production capacity claim. **Repository:** github.com/PSP-YN/Race-proof-team-formation · **Demo:** crewforge-demo-pspyn.vercel.app